

Generation of Neutrons in Giant Upward Atmospheric Discharges

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It has been shown that the efficiency of the reactions of the fusion of deuterons in the atmosphere is very low; therefore, nuclear fusion cannot be responsible for the generation of neutrons by lightning. The generation of neutrons in thunderstorm fields is attributed to photonuclear reactions in giant upward atmospheric discharges over thunderclouds.

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The hypothesis of the possibility of nuclear reactions in thunderstorm fields was proposed by Wilson in 1924, and the first estimates of the neutron yield from lightning were performed on the basis of the scaling of the ${}^2\text{H}({}^2\text{H}, n){}^3\text{He}$ fusion reaction in laboratory discharges (see references in [1]). Direct measurements of the enhancement of the neutron flux in the troposphere that accompanies thunderstorm activity were reported in [2–5]. A strong excess of the neutron flux in a thunderstorm atmosphere over the cosmic background was detected in the very fine mountain experiment [3]. The statistically significant single events in which the enhancement of the neutron flux during a thunderstorm was detected almost at the sea level were reported in [4, 5]. The results reported in [3–5] were treated as a consequence of the ${}^2\text{H}({}^2\text{H}, n){}^3\text{He}$ fusion reaction in a lightning channel.

The nuclear reactions generating neutrons could be a key to understanding the lightning mechanism. If the neutrons are generated by a stepped leader stroke, neutron data would help to understand its mechanism. If the reactions proceed in the return shock channel, which is considered as well studied at the microsecond scale, neutron data would make it possible to acquire new information on the local temperatures or local electric field strengths in dependence on which the mechanism, thermal or accelerating, is responsible for the nuclear reactions. In this work, it is shown that nuclear fusion in a thunderstorm atmosphere is certainly impossible and that the neutrons are generated in photonuclear reactions that involve relativistic outgoing electrons and proceed outside lightning channels in giant upward atmospheric discharges developed over thunderclouds in volumes up to 1000 km^3 as a result of the breakdown of the stratosphere (see [6] and references therein).

YIELD OF THE ${}^2\text{H}({}^2\text{H}, n){}^3\text{He}$ REACTION IN THE LIGHTNING CHANNEL

The expected neutron yield from the ${}^2\text{H}({}^2\text{H}, n){}^3\text{He}$ fusion reaction can be estimated by the formula

$$N_n \approx N_L P [\text{H}_2\text{O}] 2[\%] S_{\text{ch}} l_{\text{ch}} n_{\text{ion}} \Delta t \times \int_{\epsilon_{\text{fus}}}^{\infty} v_{\text{ion}} f(\epsilon_{\text{ion}}, T) \sigma_{\text{fus}}(\epsilon_{\text{ion}}) d\epsilon_{\text{ion}}. \quad (1)$$

Here, N_L is the concentration of molecules in air under normal conditions (Loschmidt number); P (atm) is the pressure; $[\text{H}_2\text{O}]$ and $[\%]$ are the relative concentration of water vapor in the thunderstorm atmosphere and deuterium molecules in natural water, respectively; ϵ_{ion} , v_{ion} , and n_{ion} are the kinetic energy, velocity, and concentration of deuterium ions, respectively; S_{ch} and l_{ch} are the cross-section area and length of the lightning channel, respectively; Δt is the existence time of strong-field regions ($(v_{\text{ion}} \Delta t \ll l_{\text{ch}})$); $\sigma_{\text{fus}}(\epsilon_{\text{ion}})$ is the cross section for nuclear fusion reactions; and ϵ_{fus} is the minimum deuteron energy at which the fusion efficiency is sufficient in the framework of the problem under consideration.

The elementary kinetic equation $eE \partial f / \partial \epsilon_{\text{ion}} = -N_L P < \sigma_t > f$, where the collision term includes only charge exchange as the most efficient process of the interaction of deuterium ions with molecules in air, provides the unity-normalized energy distribution function of deuterons $f(\epsilon_{\text{ion}})$ in the field with the strength E :

$$f(\epsilon_{\text{ion}}, T) = T^{-1} \exp(-\epsilon_{\text{ion}}/T). \quad (2)$$

Here, $T = eE/N_L P < \sigma_t >$, where $< \sigma_t >$ is the average charge-exchange cross section. Since, according to the Gamow formula for σ_{fus} , the rate $\sigma_{\text{fus}}(\epsilon_{\text{ion}}) v_{\text{ion}} \sim$

$(\epsilon_{\text{ion}})^{-0.5} \exp(-\text{const}/\sqrt{\epsilon_{\text{ion}}})$ of the ${}^2\text{H}({}^2\text{H}, n){}^3\text{He}$ reaction is a smoother function of ϵ_{ion} than function (2), the neutron yield is estimated as

$$N_n \approx N_L P[\text{H}_2\text{O}] 2[\%] S_{\text{ch}} l_{\text{ch}} n_{\text{ion}} \Delta t \times \langle v_{\text{ion}} \sigma_{\text{fus}}(\epsilon_{\text{ion}}) \rangle \exp(-\epsilon_{\text{fus}}/T). \quad (3)$$

Here, $T \approx 1.67$ eV for the maximum detected strength of the thunderstorm field $E \approx 1$ MV/m [7] (according to new measurements, $E < 100$ kV/m [8]), $N_L P \approx 10^{25} \text{ m}^{-3}$ (corresponds to an altitude of 7 km), and $\langle \sigma_t \rangle = 6 \times 10^{-20} \text{ m}^2$ (taken to be equal to σ_t for the energy $\epsilon_{\text{ion}} = 50$ eV [9]). In the return shock of the lightning, the temperature $T \approx 3$ eV was measured [7]. Even for $T \approx 10$ eV at $\epsilon_{\text{fus}} = 1.7\text{--}6.6$ keV [10], when the cross section for the ${}^2\text{H}({}^2\text{H}, n){}^3\text{He}$ reaction is very small ($\sigma_{\text{fus}} = 10^{-36}\text{--}10^{-32} \text{ m}^2$ [10]), the absolute value of the exponential argument in Eq. (3) is equal to 170–660; i.e., it is so large that $N_n \ll 1$ for reasonable values of the geophysical quantities. In particular, for $l_{\text{ch}} = 1\text{--}10$ km, $S_{\text{ch}} \approx 1\text{--}10 \text{ cm}^2$ [7], $[\text{H}_2\text{O}] \sim 1.65\%$ (according to the thickness of the “sedimentary water” layer [11]), $[\%] = 0.015\%$ [11], $\Delta t \sim 50 \mu\text{s}$ (typical duration of the return shock [3, 7]), $\sigma_{\text{fus}}(6.6 \text{ keV}) = 10^{-32} \text{ m}^2$, and $v_{\text{ion}} \sim 10^6 \text{ m/s}$ (corresponds to $\epsilon_{\text{ion}} \approx 6.6$ keV), the pre-exponential factor is much smaller than 10^{10} even for the complete ionization of all deuterium atoms in the channel, when $n_{\text{ion}} = N_L P[\text{H}_2\text{O}] 2[\%]$, which is absolutely unreal. Hence, $N_n \ll 10^{-64}$. Variation of quantities in Eq. (3) in wide ranges and the inclusion of the contribution to fusion from fast deuterium atoms that appear in the charge exchange processes and that are strongly decelerated in the dense atmosphere do not change the character of the estimate.

In order to finally demonstrate the impossibility of the ${}^2\text{H}({}^2\text{H}, n){}^3\text{He}$ reactions, it is worth noting that, even for $E = 3$ MV/m, which is the breakdown strength of the homogeneous field at $N = N_L$, the absolute value of the exponential argument in the same ϵ_{fus} range is equal to 60–220. This estimate demonstrates the impossibility of the ${}^2\text{H}({}^2\text{H}, n){}^3\text{He}$ reactions in hypothetical double layers in the return shock channel [4], because 3 MV/m is the limit value for relatively slow processes of the generation of the thunderstorm field.

NEUTRON GENERATION IN PHOTONUCLEAR REACTIONS IN UPWARD ATMOSPHERIC DISCHARGES

Upward atmospheric discharges are developed as a result of the generation of avalanches of relativistic outgoing electrons, which are initiated by cosmic rays, and are accompanied by pulses of hard gamma rays [12, 13] (see references cited in [6, 14, 15]). Since the measured gamma spectrum extends up to 20 MeV and the calculated mean energy of the photons of the bremsstrahlung

of the avalanches of relativistic outgoing electrons is ~ 1 MeV [14], the generation of neutrons is possible in the (γ, n) reactions outside the channel of a lightning, which only triggers the upward atmospheric discharge mechanism.

Let us use the upward atmospheric discharge model, according to which the electrons of the relativistic avalanches near a certain altitude H move horizontally owing to the effect of the vertical component of the geomagnetic field [15]. In the approximation of the isotropic point source of the bremsstrahlung, the number of photoneutrons generated per discharge can be estimated as

$$N_n = 2N_L P(H) \int_{H-h_{\text{char}}}^{H+h_{\text{char}}} e^{-x/h_{\text{char}}} dx \int_0^{\Delta t_\gamma} \frac{dN_\gamma(\delta)}{dt} \times N_e(\delta, t) dt \int_{\epsilon_{\text{th}}(\gamma, 1n)}^{\infty} f_\gamma(\delta, \epsilon_\gamma) \sigma(\gamma, Sn) d\epsilon_\gamma \quad (4)$$

$$\approx 2N_L \exp\left(-\frac{2H}{h_{\text{char}}}\right) h_{\text{char}} \left(e - \frac{1}{e}\right) N_e(\delta) \times \frac{dN_\gamma}{dt} \Delta t_\gamma \sigma_{\text{yield}}(\epsilon_{\gamma, \text{max}}) \langle f_\gamma(\delta, \epsilon_{\text{th}}) \rangle.$$

Here, the exponential altitude dependence of the air density is accepted with a characteristic scale of $h_{\text{char}} \approx 7.1$ km, $dN_\gamma(\delta)/dt$ is the photon-emission rate, Δt_γ is the duration of the pulses of atmospheric gamma rays detected on orbital spacecrafts [12, 13], $N_e(\delta)$ is the number of relativistic outgoing electrons in air at the overvoltage $\delta = eE/(F_{\text{min}}P)$ with respect to the electron friction force minimum $F_{\text{min}} = 218 \text{ keV/m atm}$, $f_\gamma(\delta, \epsilon_\gamma)$ is the unity-normalized distribution of photons over the energy ϵ_γ , $\sigma(\gamma, Sn) = \sum_i i \sigma(\gamma, in) + \nu \sigma(\gamma, f)$, $\sigma(\gamma, in)$ is the cross section for the (γ, in) reaction with the yield of i neutrons, $\sigma(\gamma, f)$ is the cross section for the nuclear photofission with the yield of ν neutrons, $\sigma_{\text{yield}}(\epsilon_{\gamma, \text{max}}) = \int_{\epsilon_{\text{th}}(\gamma, 1n)}^{\epsilon_{\gamma, \text{max}}} \sigma(\gamma, Sn) d\epsilon$ is the cross section for the neutron yield in all the photonuclear reactions, $\epsilon_{\text{th}}(\gamma, 1n)$ is the threshold of the $(\gamma, 1n)$ reaction, and $\epsilon_{\gamma, \text{max}}$ is the maximum energy to which data on the cross section exist [16]. Within the accuracy of our estimates, it is sufficient to treat the atmosphere as consisting of ${}^{14}\text{N}$ nuclei for which $\epsilon_{\text{th}}(\gamma, 1n) = 10.55 \text{ MeV}$, $\epsilon_{\gamma, \text{max}} = 29.5 \text{ MeV}$, and $\sigma_{\text{yield}} = 98.8 \times 10^{-31} \text{ MeV m}^2$ [16]. Setting $\delta = 4$ (altitude $H = 30$ km) [15], and $\Delta t_\gamma \approx 1 \text{ ms}$ [12, 13] (this

time includes the series of the generation of avalanches of relativistic outgoing electrons [15]),

$$\langle f_{\gamma}(\delta, \varepsilon_{\text{th}}) \rangle > \frac{1}{\varepsilon_{\gamma, \text{max}} - \varepsilon_{\text{th}}(\gamma, n)} \int_{\varepsilon_{\text{th}}(\gamma, n)}^{\varepsilon_{\gamma, \text{max}}} f_{\gamma}(\delta, \varepsilon_{\gamma}) d\varepsilon_{\gamma} \\ \approx 5 \times 10^{-4} \text{ 1/MeV},$$

$dN_{\gamma}(\delta)/dt \approx 10^7 \text{ s}^{-1} \text{ atm electron}$ [14], and $N_e(\delta) = 10^{17}$, which corresponds to $H \approx 30 \text{ km}$ [15] and is sufficient for the generation of the number of hard photons detected in [12]; one obtains the number $N_n \approx 10^{15}$ of neutrons generated in one upward atmospheric discharge.

NEUTRON DELAY TIME

In [3], the delay times t_{del} of the arrival of the neutrons on a counter with respect to the electromagnetic pulse were in the range $10 \mu\text{s}$ – 0.1 s , and the number of neutron events decreases with increasing t_{del} . Under the assumption that t_{del} is the time of the rectilinear motion [3] of the neutron with the energy $\varepsilon_n = 1$ – 10 MeV , for $t_{\text{del}} \approx 10 \mu\text{s}$ and 0.1 s , the sources are at distances of hundreds of meters and thousands of kilometers from the monitor, respectively. Large t_{del} values are explained as follows [3].

(i) The first lightning discharge triggers the counter, and the neutrons are generated by the next flashes.

(ii) The $^{12}\text{N}(^2\text{H}, n)^{13}\text{C}$ and $^{14}\text{N}(^2\text{H}, n)^{15}\text{O}$ reactions generate neutrons with lower energies than those in the $^2\text{H}(^2\text{H}, n)^3\text{He}$ reaction.

(iii) Neutrons walk in the air before arriving at the counter.

In the framework of the photonuclear mechanism, the wide t_{del} range is explained by two causes. First, the neutrons are generated in a large volume on the photon propagation path from large altitudes, where upward atmospheric discharges are developed, to the ground, in the ground, and in the monitor substance. Second, in contrast to nuclear-fusion neutrons, the photoneutrons are distributed over a wide energy range $\varepsilon_n = [0; \varepsilon_{\gamma} - \varepsilon_{\text{th}}(\gamma, 1n)]$. Large t_{del} values are detected when the source of the first neutron with a sufficiently high energy is at a large distance from the monitor. Small t_{del} values are due to the generation of neutrons by photons that reach the surface troposphere layers. However, since only the photon propagation time from altitudes at which upward atmospheric discharges are developed is close to $100 \mu\text{s}$, the problem of the coordination of the mechanism of neutron generation by gamma rays from upward atmospheric discharges with $t_{\text{del}} \leq 100 \mu\text{s}$ is naturally solved if the counter for events with $t_{\text{del}} \leq 100 \mu\text{s}$ is triggered by the electromagnetic pulse of upward atmospheric discharges.

Shah et al. [3] mentioned an event in which 33 neutrons were detected when lightning struck a tree at a distance of 400 m from the monitor. The time $t_{\text{del}} = 72 \text{ ms}$ [3] corresponding to this event strongly contradicts the nuclear fusion mechanism in the lightning channel, but it is naturally explained by the photonuclear mechanism of neutron generation by gamma radiation from upward atmospheric discharges.

CONCLUSIONS

The neutron yield in the photonuclear reactions that accompany gamma radiation from upward atmospheric discharges is estimated as $\sim 10^{15}$, whereas nuclear fusion is absolutely impossible under the conditions that, according to the current representations, are realized in the lightning channel. The number of outgoing electrons, 10^{17} , that is obtained on the basis of the upward atmospheric discharge mechanism proposed by Gurevich, Milikh, and Roussel-Dupré (see [6] and references cited therein) and that is sufficient for satisfactory explanation of the gamma bursts of atmospheric origin [12, 13] leads to the generation of 10^{15} neutrons in the (γ, n) reactions. Comparison with natural observation data requires a numerical simulation including the self-consistent transport of gamma radiation and neutrons over large distances in the formulation corresponding to the experimental conditions.

The existence of a connection between the enhancement of the atmospheric neutron flux that was detected in [3] and upward atmospheric discharges would be serious evidence in favor of the upward atmospheric discharge mechanism involving relativistic outgoing electrons. The absence of such a connection would mean either that the correlation of multineutron events with lightning that was revealed in [3] is accidental, which is doubtful, because these events were carefully selected, or that the lightning involves local processes with characteristic times much shorter than a microsecond that can be responsible for the enhancement of the neutron flux.

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